LAPORAN PRAKTIKUM

POST TEST 7

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Muhammad Athaillah Cordova (2509106055)

Kelas B1 ‘25

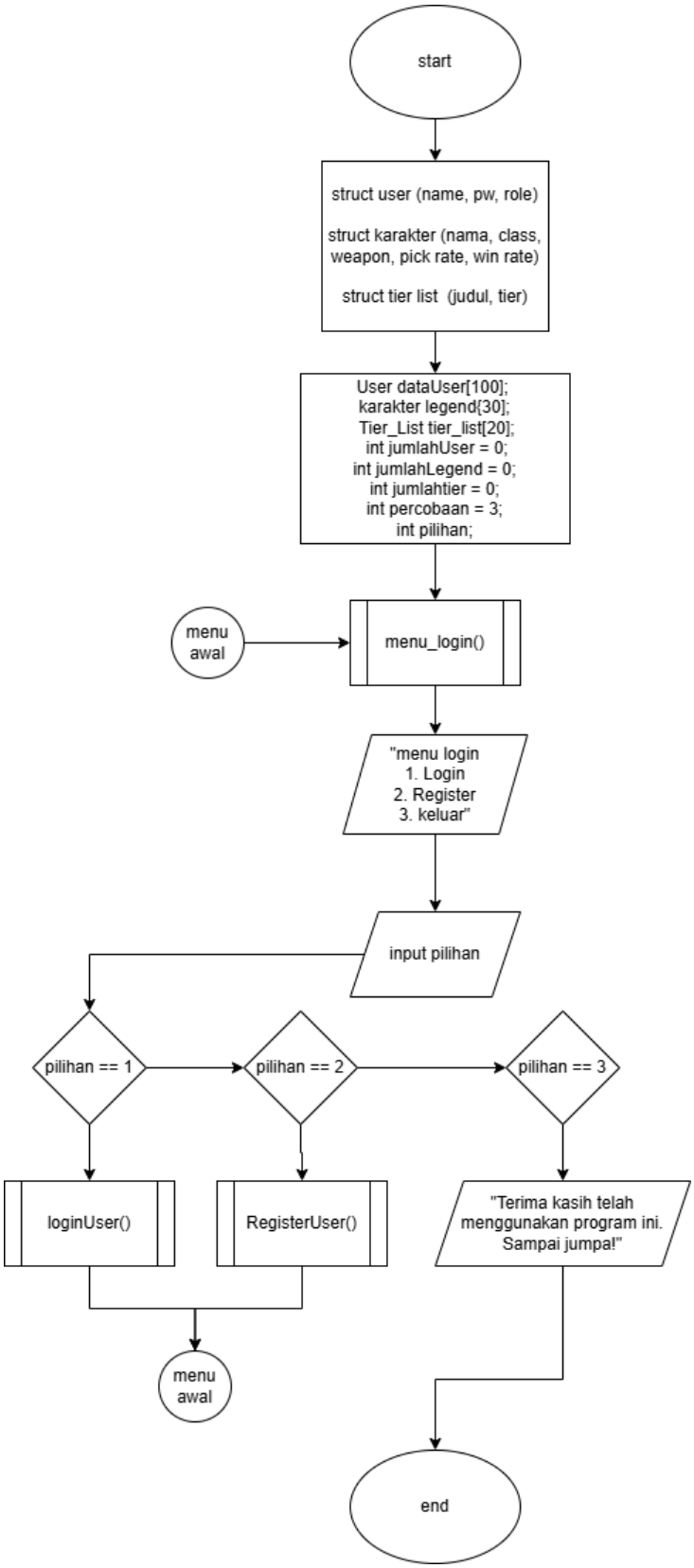
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

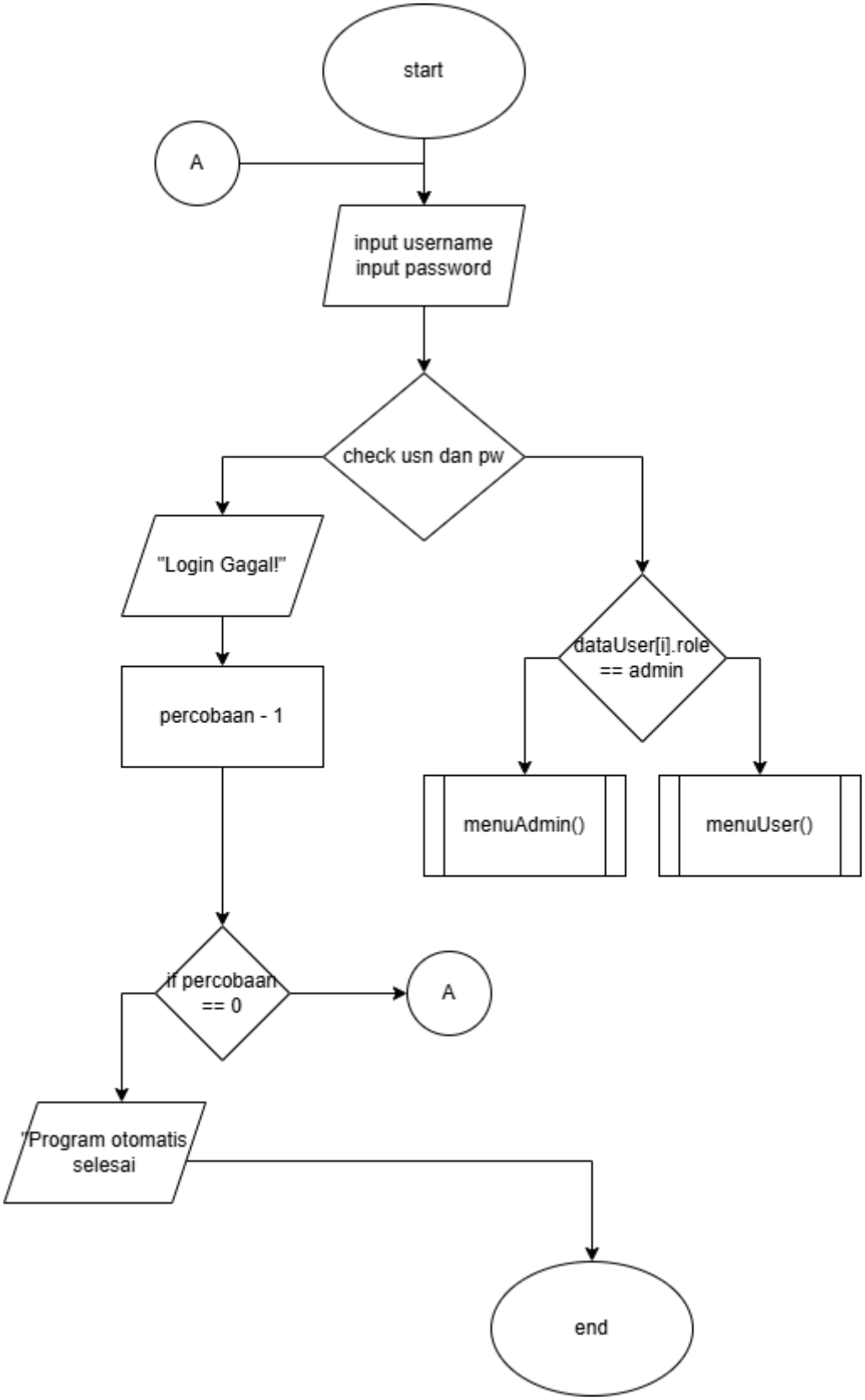
SAMARINDA

2025

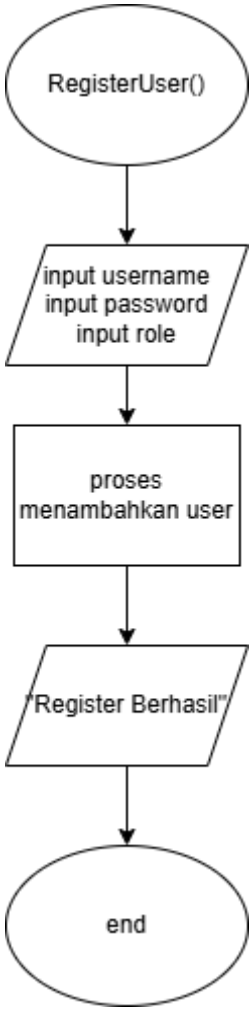
1. Flowchart



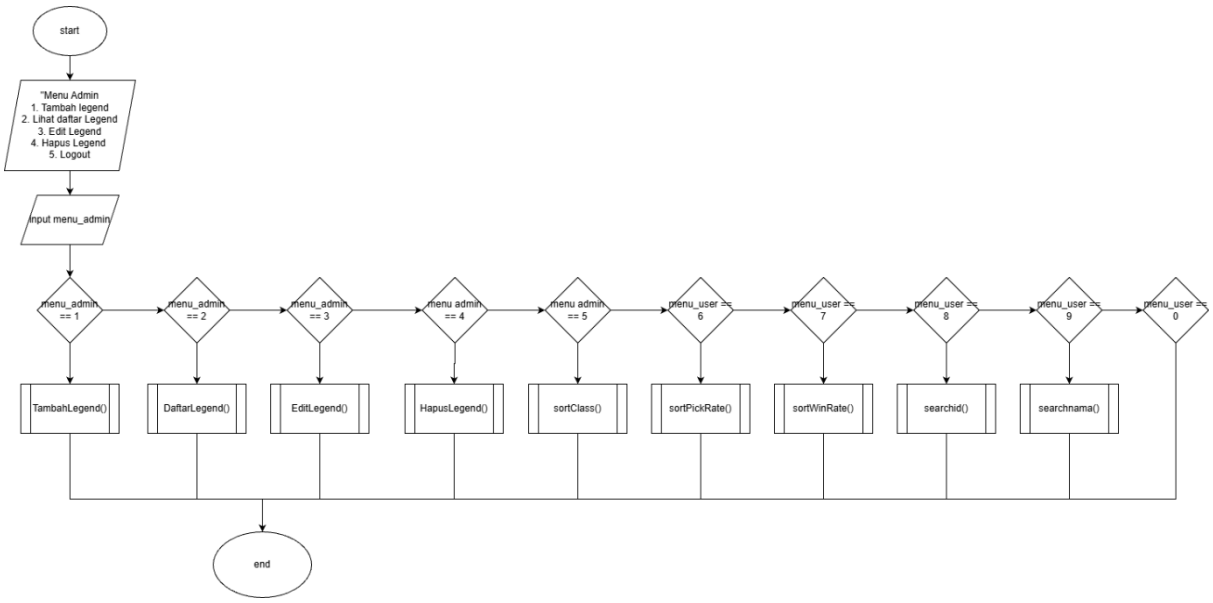
Gambar 1.1 Flowchart Menu Utama



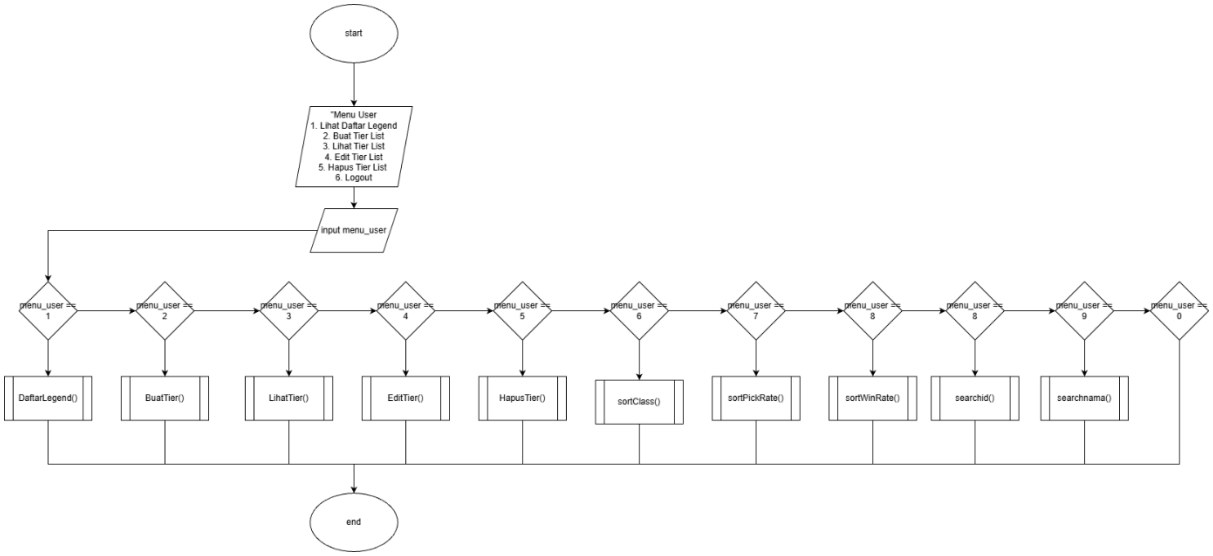
Gambar 1.2 Flowchart Fungsi loginUser



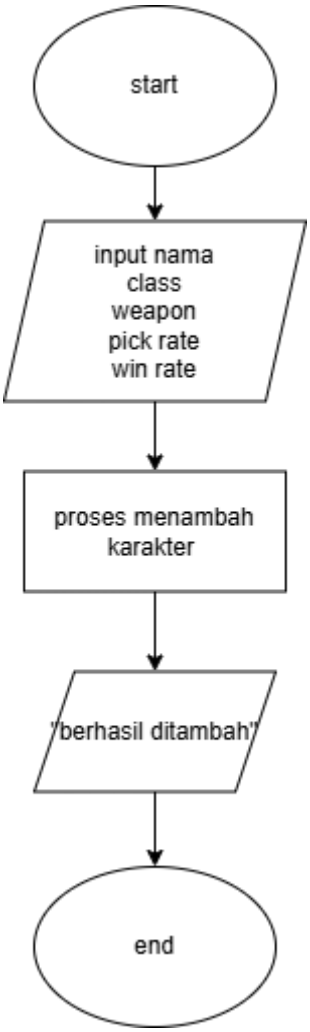
Gambar 1.3 Flowchart Fungsi RegisterUser



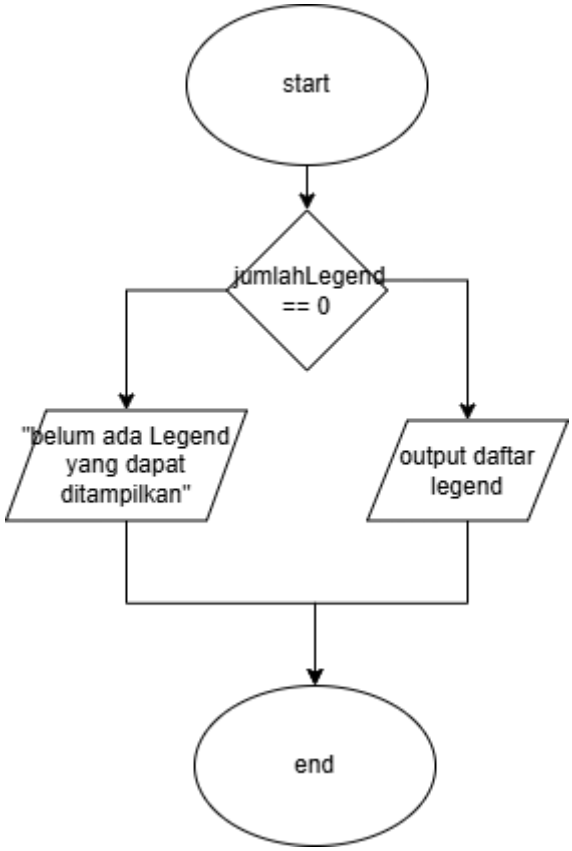
Gambar 1.4 Flowchart Fungsi menuAdmin



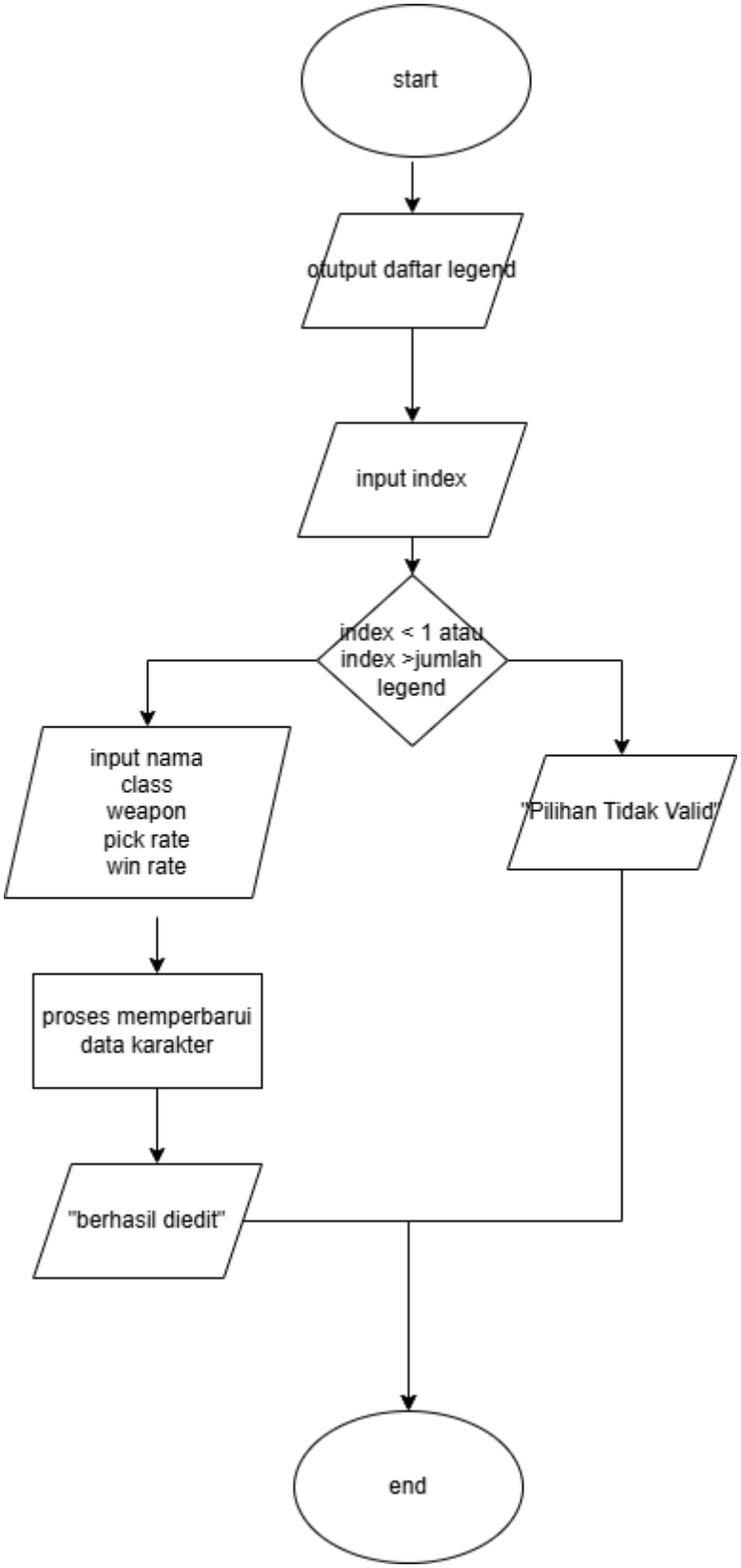
Gambar 1.5 Flowchart Fungsi menuUser



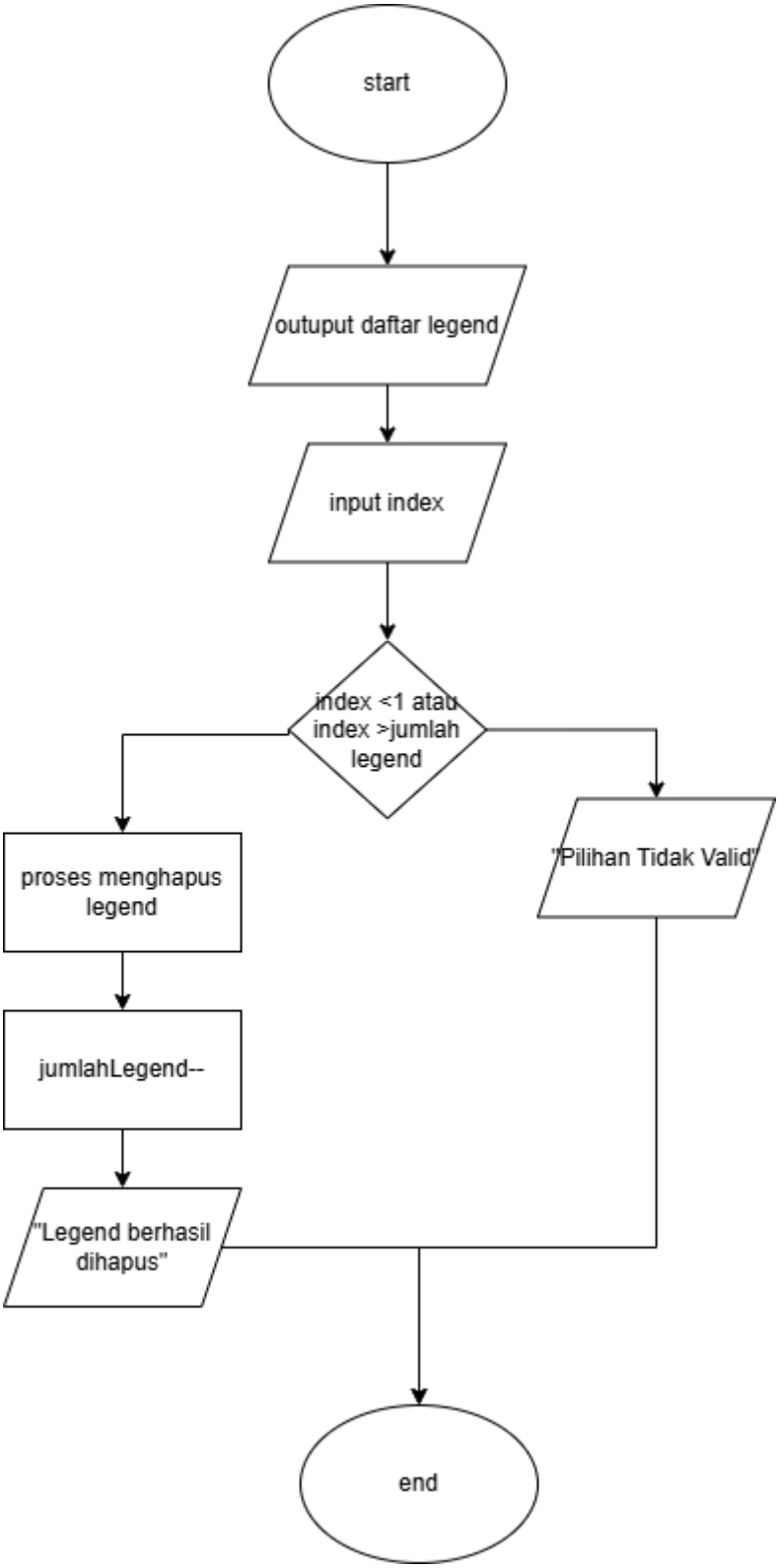
Gambar 1.6 Flowchart Fungsi TambahLegend



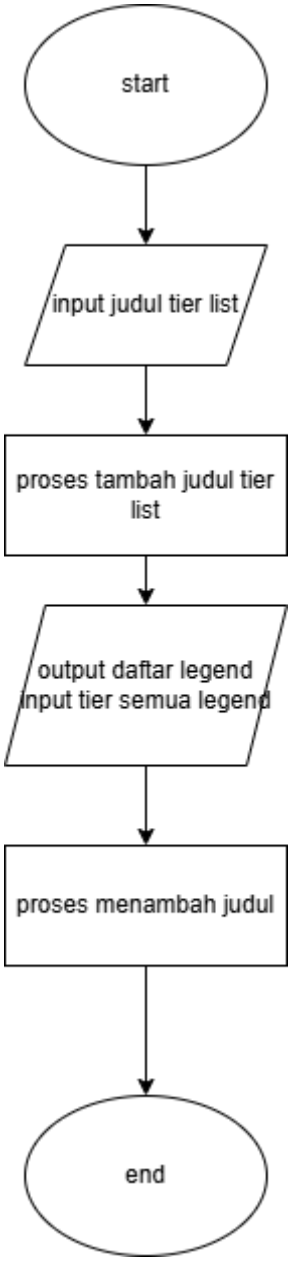
Gambar 1.7 Flowchart Fungsi TampilLegend



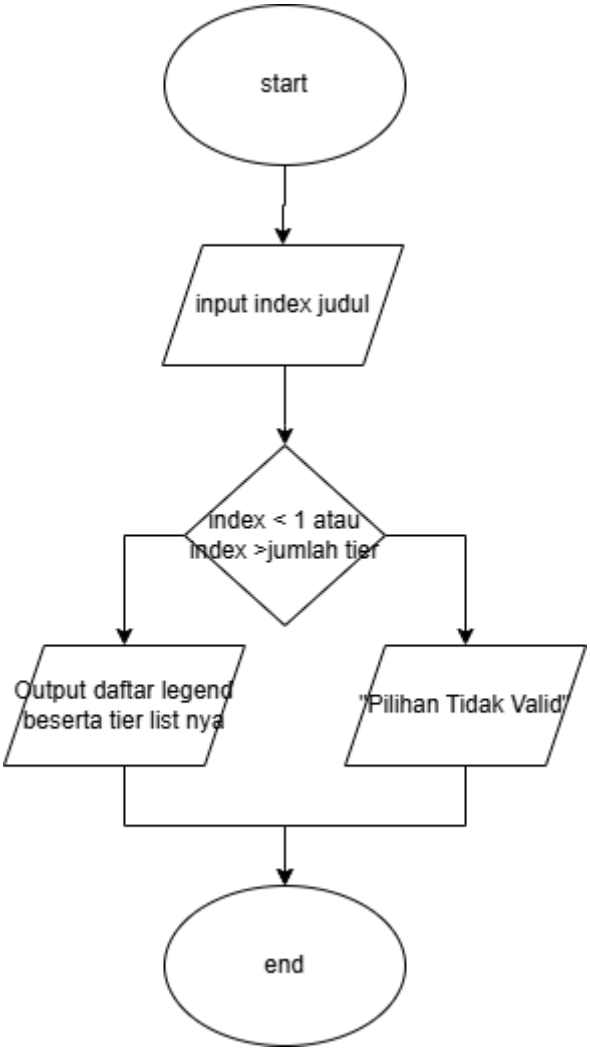
Gambar 1.8 Flowchart Fungsi EditLegend



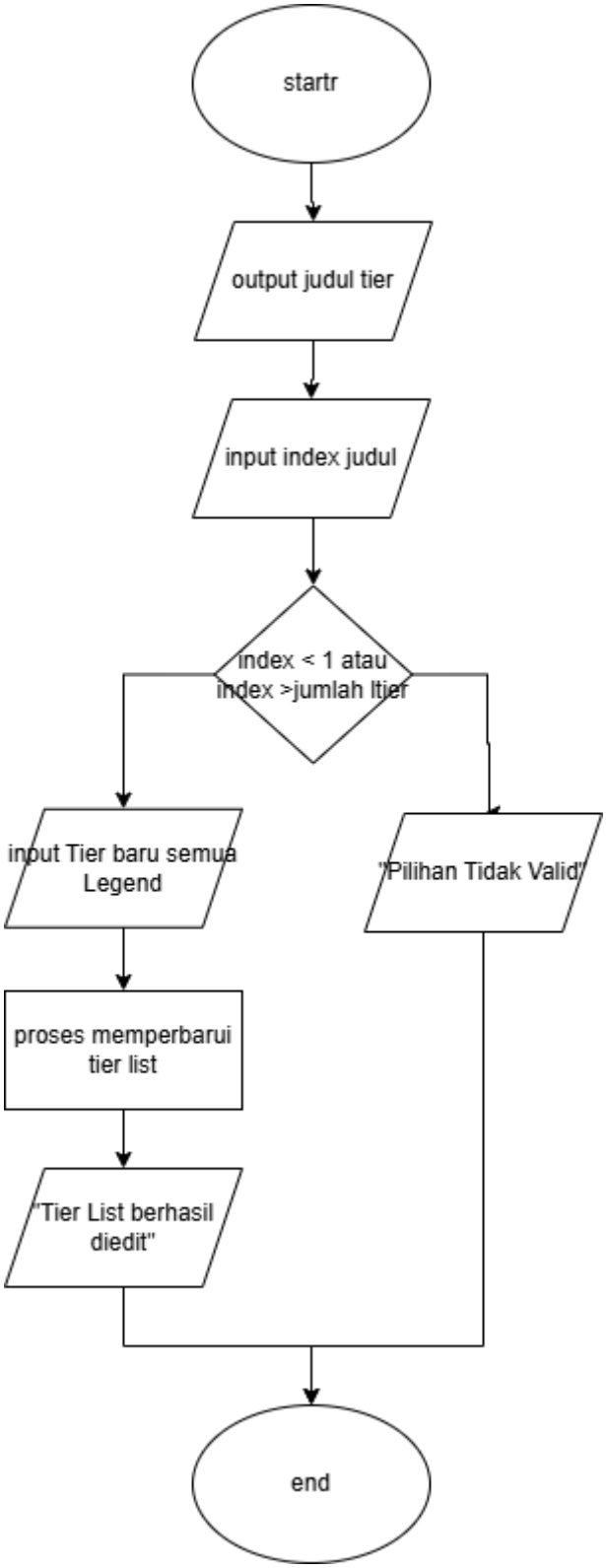
Gambar 1.9 Flowchart Fungsi hapusLegend



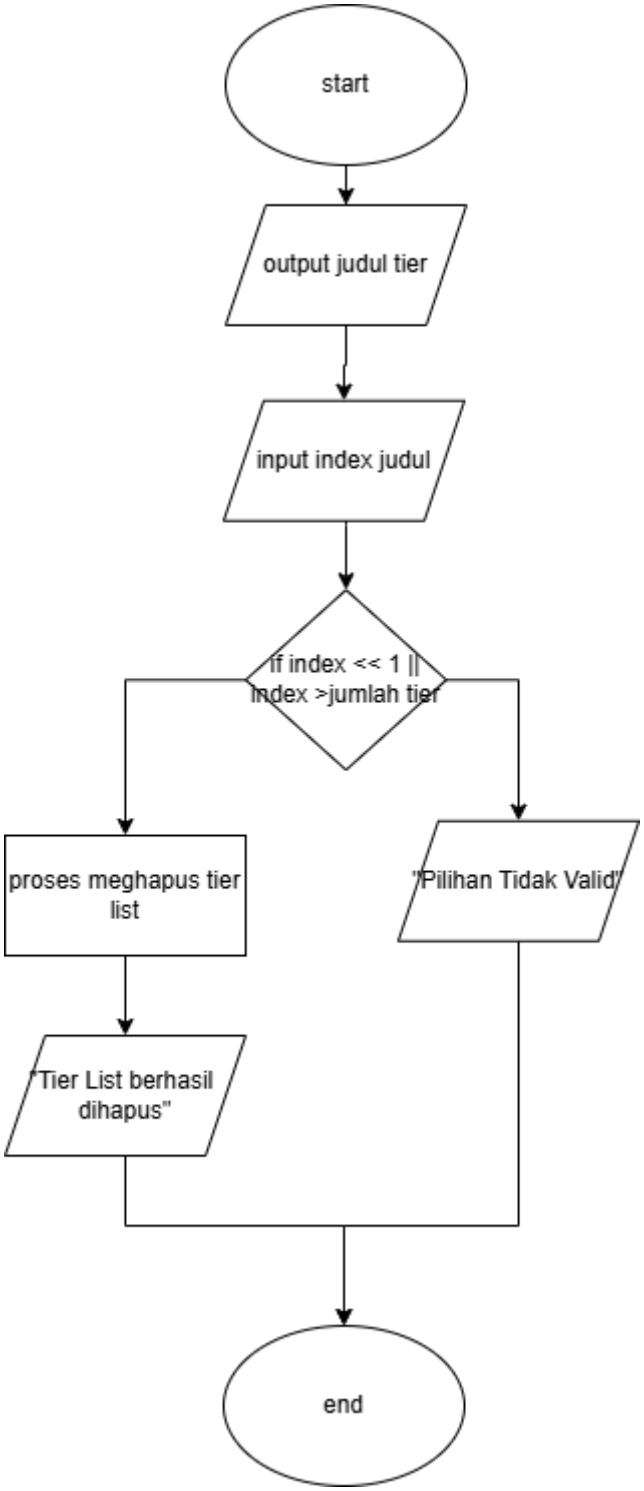
Gambar 1.10 Flowchart Fungsi BuatTier



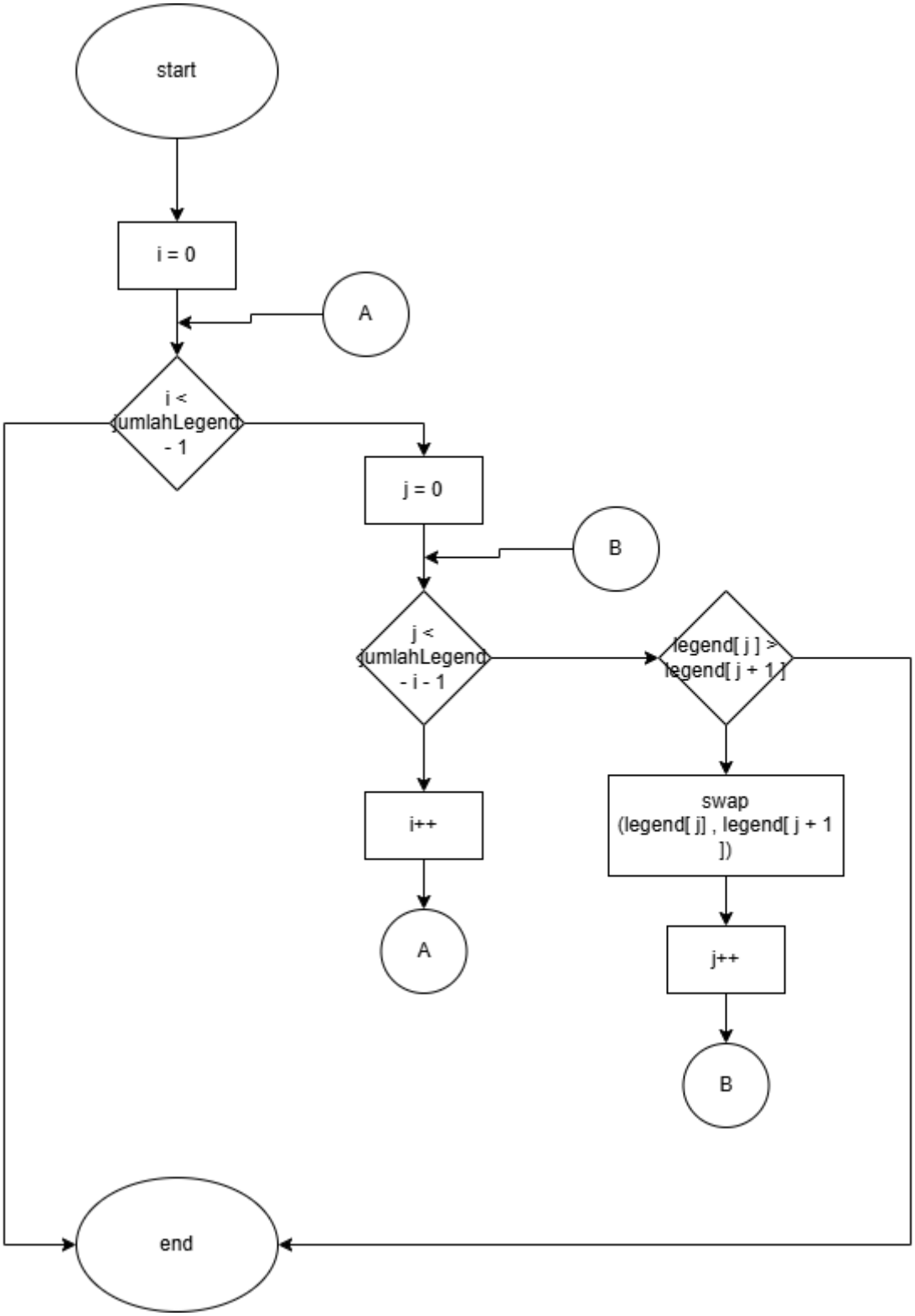
Gambar 1.11 Flowchart Fungsi LihatTier



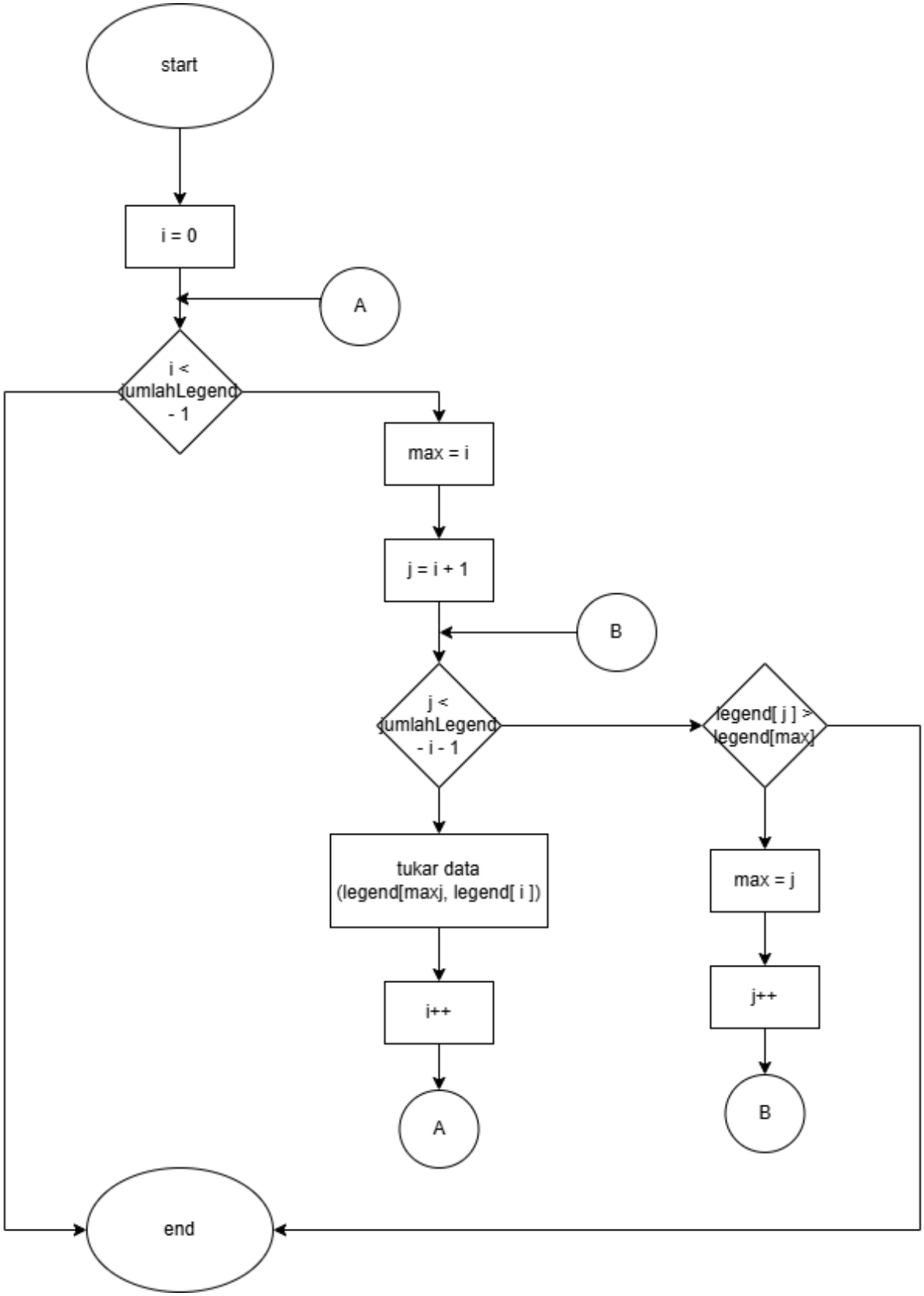
Gambar 1.12 Flowchart Fungsi EditTier



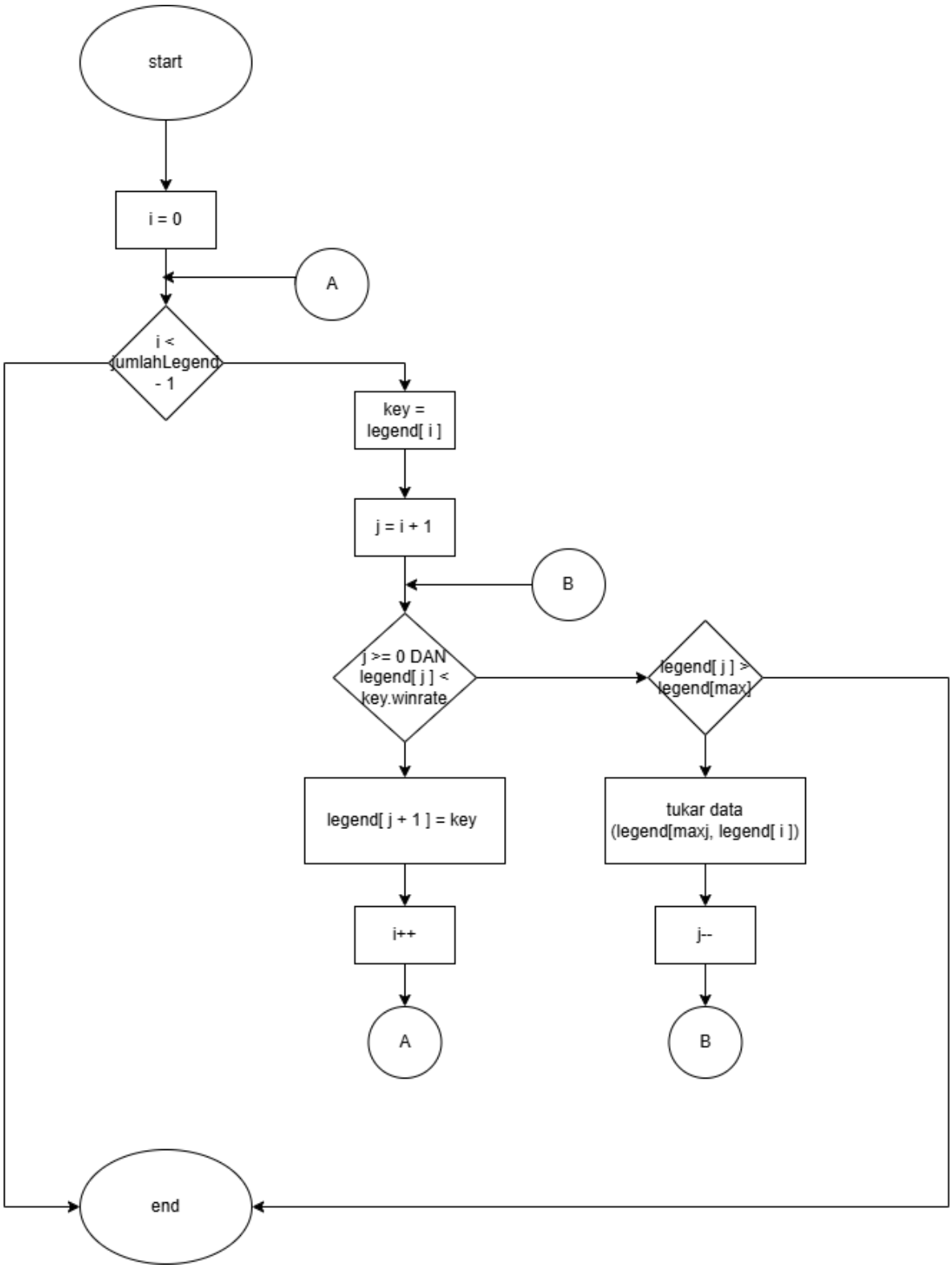
Gambar 1.13 Flowchart Fungsi hapusTier



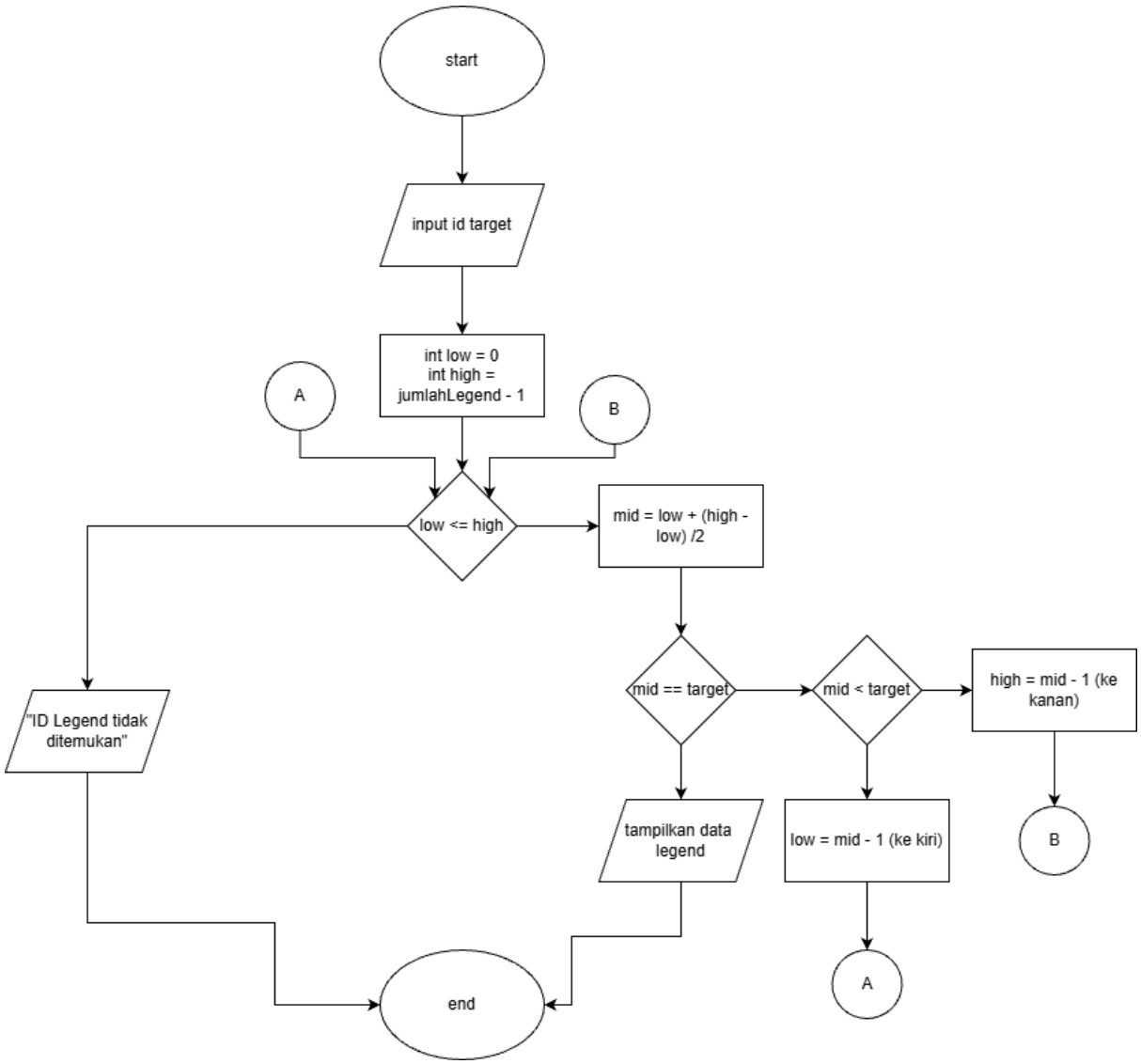
Gambar 1.14 Flowchart Fungsi sortClass



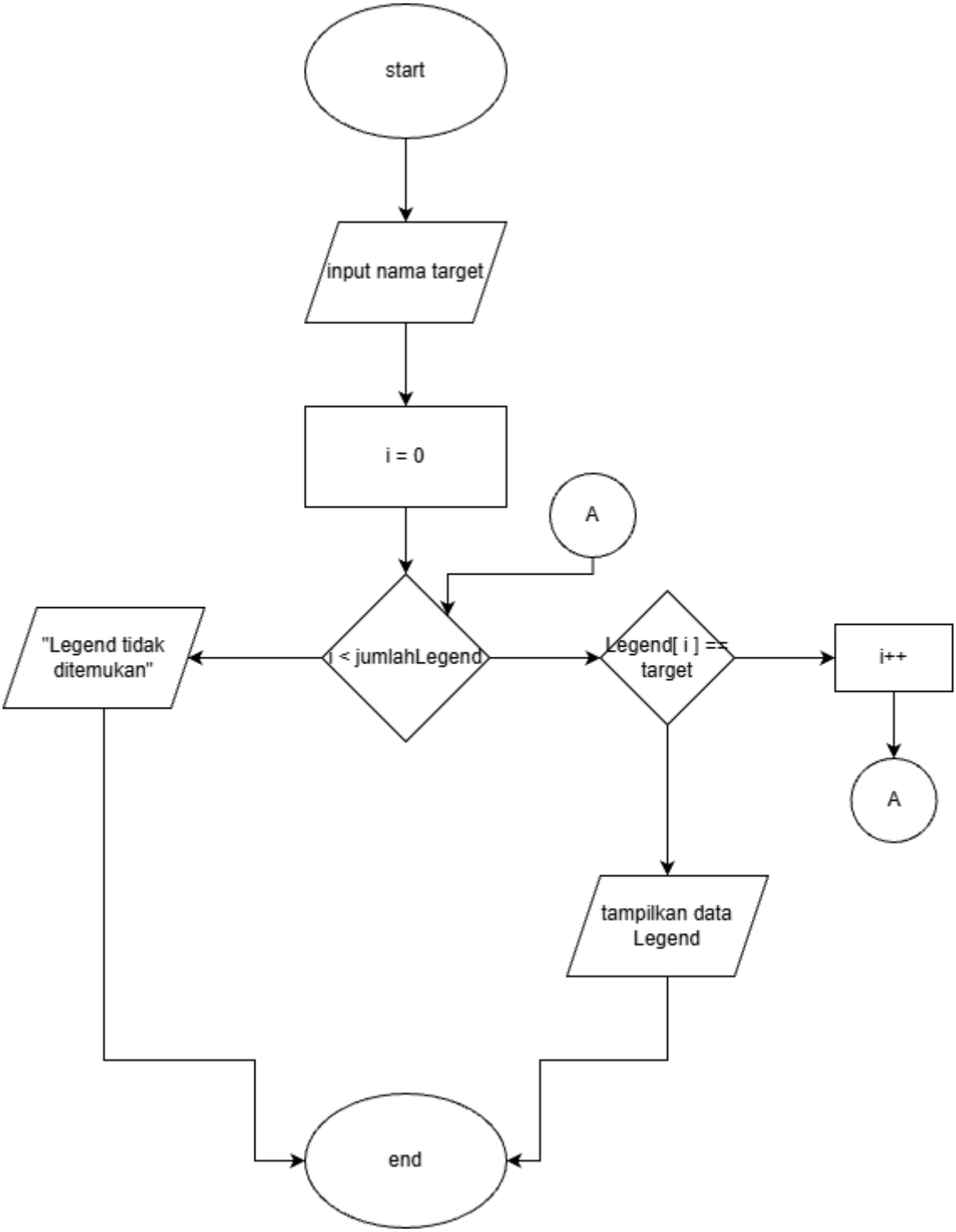
Gambar 1.15 Flowchart Fungsi sortPickRate



Gambar 1.16 Flowchart Fungsi sortWinRate



Gambar 1.17 Flowchart Fungsi searchid



Gambar 1.18 Flowchart Fungsi searchnama

2. Deskripsi Singkat Program

Program ini bertemakan sistem data Karakter Game Apex Legend, di program ini terdapat dua role user saat login yakni admin dan user. Di menu login, kita bisa melakuakn registrasi terlebih dahulu jika belum memilki akun, setelah registrasi akan kembali ke halaman menu lagi lau bisa langsung login, tetapi ada batas untuk bisa melakukan login sebanyak 3 kali jika 3 kali salah maka program otomatis berhenti/selesai. Saat login program akan langsung mengetahui role apa akun tersebut.

Lanjut Kebagian CRUD, role admin Memiliki akses untuk melakukan CRUD pada karakter yang tersedia. Untuk user, user Bisa melakukan CRUD berupa Tier list dari karakter yang tersedia. Saat Membuat tier list user akan diminta memberi judul tierlist lalu user memberi tier pada karakter yang tersedia umtuk edit, user bisa mengedit ulang tier seluruh karakter, dan untuk hapus, user bisa menghapus tier list yang sudah dibaut.

Terdapat Fitur baru di program yakni sorting legend. Ada tiga sorting yang dapat dilakukan oleh admin maupun user dimenu masing-masing. Sorting legend Berdasarkan Class dengan diurutkan secara Ascending (A-Z), lalu ada sortng legend berdasarkan Pick Rate dengan diurutkan secara Descending (Terbesar-Terkecil), dan yang terakhir ada sorting berdasarkan Win Rate yang diurutkan secara Descending (Terbesar-Terkecil).

Menambha Fitur baru yakni Searching. Terdapat dua searching yang bisa dilakukan, diantaranya adalah searching berdasarkan id, dan searching berdasarkan nama, searcing id menggunakan binary search sedangka, search nama menggunakan linear search.

3. Source Code

```
const int MAX_USER = 100;
const int MAX_LEGENDS = 30;
const int MAXLIST = 20;

struct User {
    string usn;
    string pw;
    string role;
};

struct karakter {
    int id;
    string nama;
    string classs;
    string best_weapon;
    float pick_rate;
    float win_rate;
};
struct Tier_List {
    string judul;
    char tier[MAX_LEGENDS];
};

int main() {

    User dataUser[MAX_USER];
    karakter legend[MAX_LEGENDS];
    Tier_List tier_list[MAXLIST];
    int jumlahUser = 1;
    int jumlahLegend = 11;
    int jumlahtier = 0;;
    int percobaan = 3;
    int pilihan;
    int indexUser = -1 ;
    dataUser[0] = {"admin", "admin123", "admin"};

    legend[0] = {101, "Alter", "Skirmisher", "SMG", 14.5, 52};
    legend[1] = {102, "Ash", "Skirmisher", "SMG", 0.4, 49.4};
    legend[2] = {103, "Bangalore", "Assault", "SMG", 7.4, 49.1};
    legend[3] = {104, "Fuse", "Assault", "SMG", 1.1, 48.8};
    legend[4] = {105, "Horizon", "Skirmisher", "AR", 3.6, 47.5};
    legend[5] = {106, "Loba", "Support", "AR", 2.8, 51.7};
    legend[6] = {107, "Newcastle", "Support", "SG", 0.6, 48.1};
    legend[7] = {108, "Seer", "Recon", "AR", 1.3, 48.7};
    legend[8] = {109, "valkyrie", "Recon", "AR", 9.2, 54.4};
    legend[9] = {1010,"Wattson", "Controller", "SMG", 0.2, 51.7};
    legend[10] = {1011,"Wraith", "Skirmisher", "SMG", 0.6, 47.5};

```

```
void RegisterUser(User dataUser[], int *jumlahUser) {
    if (*jumlahUser >= MAX_USER) {
    cout << "Maaf, kapasitas pengguna sudah penuh.\n";
    pause();
    return;
    }

    string usn, pw, role;
    cout << "Masukkan Username: ";
    cin >> usn;
    cout << "Masukkan Password: ";
    cin >> pw;
    cout << "Masukkan role (admin/user): ";
    cin >> role;

    if (role != "admin" && role != "user") {
        cout << "Role tidak valid. Harus 'admin' atau 'user'.\n";
        pause();
        return;
    }
    dataUser[*jumlahUser].usn = usn;
    dataUser[*jumlahUser].pw = pw;
    dataUser[*jumlahUser].role = role;
    (*jumlahUser)++;
    cout << "\nSelamat Registrasi berhasil! Silahkan Login\n";
    pause();
}

int LoginUser(User dataUser[], int *jumlahUser){

    string usn, pw;

    cout << "Username: ";
    cin >> usn;

    cout << "Password: ";
    cin >> pw;

    for(int i = 0; i < *jumlahUser; i++){
        if(dataUser[i].usn == usn && dataUser[i].pw == pw){
            return i;
        }
    }

    return -1;
}

void menu_login() {
    cout << "\n===== MENU LOGIN =====\n";
    cout << "1. Login\n";
    cout << "2. Register\n";
    cout << "0. Keluar\n";
    cout << "Pilih menu: ";
}

void menuAdmin() {
    cout << "\n===== MENU ADMIN =====\n";
    cout << "1. Tambah Legend\n";
    cout << "2. Lihat Daftar legend\n";
    cout << "3. Edit Legend\n";
    cout << "4. Hapus Legend\n";
    cout << "5. Urutkan Class Legend\n";
    cout << "6. Urutkan Pick Rate Legend\n";
    cout << "7. Urutkan Win Rate Legend\n";
    cout << "8. Cari ID Legend\n";
    cout << "9. Cari Nama Legend\n";
    cout << "0. Logout\n";
    cout << "Pilih menu: ";
}

void menuUser() {
    cout << "\n===== MENU USER =====\n";
    cout << "1. Lihat Daftar Legend\n";
    cout << "2. Urutkan Class Legend\n";
    cout << "3. Urutkan Pick Rate Legend\n";
    cout << "4. Urutkan Win Rate Legend\n";
    cout << "5. Cari ID Legend\n";
    cout << "6. Cari Nama Legend\n";
    cout << "7. Buat Tier List\n";
    cout << "8. Lihat Tier List\n";
    cout << "9. Edit Tier List\n";
    cout << "10. Hapus Tier List\n";
    cout << "0. Logout\n";
    cout << "Pilih menu: ";
}
```

```cpp
void TambahLegend(karakter legend[], int *jumlahLegend) {
    cout << "\nTambah Legend\n";
    cout << "Nama: ";
    cin >> legend[*jumlahLegend].nama;
    cout << "Class: ";
    cin >> legend[*jumlahLegend].classs;
    cout << "Best Weapon: ";
    cin >> legend[*jumlahLegend].best_weapon;
    cout << "Pick Rate: ";
    cin >> legend[*jumlahLegend].pick_rate;
    cout << "Win Rate: ";
    cin >> legend[*jumlahLegend].win_rate;
    (*jumlahLegend)++;
    cout << "Legend berhasil ditambahkan!\n";
    pause();
}

void DaftarLegend(karakter legend[], int &jumlahLegend) {
    cout << "\nLegend\n";
    if (jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat ditampilkan.\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(10) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==========================================================\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "| " << setw(10) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(0) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(0) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==========================================================\n";
    pause();
    }
}
```

```cpp
void EditLegend(karakter legend[], int *jumlahLegend) {
    cout << "\nEdit Legend\n";
    if (*jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat diedit.\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(10) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==========================================================\n";
    for (int i = 0; i < *jumlahLegend; i++) {
        cout << "| " << setw(10) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(0) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(0) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==========================================================\n";
        int index;
        cout << "\nPilih nomor legend yang ingin diedit: ";
        cin >> index;
        if (index < 1 || index > *jumlahLegend) {
            cout << "Pilihan tidak valid.\n";
        } else {
            index--;
            cout << "\nEdit Legend: " << legend[index].nama << "\n";
            cout << "\nNama: ";
            cin >> legend[index].nama;
            cout << "Class: ";
            cin >> legend[index].classs;
            cout << "Best Weapon: ";
            cin >> legend[index].best_weapon;
            cout << "Pick Rate: ";
            cin >> legend[index].pick_rate;
            cout << "Win Rate: ";
            cin >> legend[index].win_rate;
            cout << "\nLegend berhasil diedit!\n";
            pause();
        }
    }
}

void HapusLegend(karakter legend[], int *jumlahLegend) {
    cout << "\nHapus Legend\n";
    if (*jumlahLegend == 0) {
        cout << "Belum ada legend yang dapat dihapus\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(12) << left << "Nama" << "|" << setw(12) << left << "Class" << "|" << setw(12) << left << "Best Weapon" << "|" << setw(10) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==========================================================\n";
    for (int i = 0; i < *jumlahLegend; i++) {
        cout << "| " << setw(10) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(0) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(0) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==========================================================\n";
        int index;
        cout << "\nPilih nomor legend yang ingin dihapus: ";
        cin >> index;
        if (index < 1 || index > *jumlahLegend) {
            cout << "Pilihan tidak valid.\n";
        } else {
            for (int i = index - 1; i < *jumlahLegend - 1; i++) {
                legend[i] = legend[i + 1];
            }
            (*jumlahLegend)--;
            cout << "\nLegend berhasil dihapus!\n";
            pause();
        }
    }
}
```

```cpp
void BuatTier(Tier_List tier_list[], int *jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nBuat Tier List\n";
    cout << "Masukkan judul tier list: ";
    cin >> ws;
    getline(cin, tier_list[*jumlahtier].judul);
    cout << "\nMasukkan tier setiap legend (S/A/B/C/D):\n";
    for ( int i = 0; i < jumlahLegend; i++) {
        cout << "Tier legend " << setw(5) << left << legend[i].nama << ": ";
        cin >> tier_list[*jumlahtier].tier[i];
    }
    (*jumlahtier)++;
    cout << "Tier list berhasil dibuat!\n";
    pause();
}

void LihatTIer(Tier_List tier_list[], int &jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List\n";
    if (jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat ditampilkan.\n";
        pause();
        return;
    }else {
    for (int i = 0; i < jumlahtier; i++) {
        cout << i + 1 << ". " << tier_list[i].judul << "\n";
    }
    cout << endl;
    int index_list;
    cout << "\nPilih tier list yang ingin dilihat: ";
    cin >> index_list;
    if (index_list < 1 || index_list > jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_list--;

    cout << "\n Tier List: " << tier_list[index_list].judul << "\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << setw(10) << left << legend[i].nama << " = " << setw(2)<< tier_list[index_list].tier[i] << "\n";
            }
            pause();
        }
    }
}
```

```
void EditTier(Tier_List tier_list[], int *jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List.\n";
    if (*jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat diedit.\n";
        pause();
        return;
    } else {
    for (int i = 0; i < *jumlahtier; i++) {
            cout << i + 1 << ". " << tier_list[i].judul <<"\n";
    }
    cout << endl;
    int index_edit;
    cout << "\nPilih tier list yang ingin diedit: ";
    cin >> index_edit;
    if (index_edit < 1 || index_edit > *jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_edit--;

    cout << "\nMasukkan tier baru setiap legend (S/A/B/C/D):\n";
    for (int i = 0; i < jumlahLegend; i++) {
        cout << "Masukkan Tier baru " << setw(5) << left << legend[i].nama << ": ";
        cin >> tier_list[index_edit].tier[i];
            }
        }
    }
    cout << "\nTier list berhasil diedit\n";
    pause();
}

void HapusTier(Tier_List tier_list[], int *jumlahtier, karakter legend[], int jumlahLegend){
    cout << "\nDaftar Tier List.\n";
    if (*jumlahtier == 0) {
        cout << "Belum ada tier list yang dapat dihapus.\n";
        pause();
        return;
    } else {
    for (int i = 0; i < *jumlahtier; i++) {
            cout << i + 1 << ". " << tier_list[i].judul << "\n";
    }
    int index_hapus;
    cout << "\nPilih tier list yang ingin dihapus: ";
    cin >> index_hapus;
    if (index_hapus < 1 || index_hapus > *jumlahtier) {
        cout << "Pilihan tidak valid.\n";
        pause();
        return;
    } else {
    index_hapus--;

    for (int i = index_hapus; i < *jumlahtier - 1; i++) {
        tier_list[i] = tier_list[i + 1];
            }
        }
    }
    (*jumlahtier)--;
    cout << "\nTier list berhasil dihapus\n";
    pause();
}
```

```cpp

void sortClass(karakter legend[], int jumlahLegend) {
    for(int i = 0; i < jumlahLegend - 1; i++) {
        for(int j = 0; j < jumlahLegend - i - 1; j++) {
            if(legend[j].classs > legend[j + 1].classs) {
                swap(legend[j], legend[j + 1]);
            }
        }
    }
    cout << "\nLegend berhasil diurutkan berdasarkan Class!\n";
    pause();
}

void sortPickrate(karakter legend[], int jumlahLegend) {
    for(int i = 0; i < jumlahLegend - 1; i++) {
        int max = i;
        for(int j = i + 1; j < jumlahLegend; j++) {
            if(legend[j].pick_rate > legend[max].pick_rate) {
                max = j;
            }
        }
        swap(legend[max], legend[i]);
    }
    cout << "\nLegend berhasil diurutkan berdasarkan Pick Rate tertinggi!\n";
    pause();
}

void sortWinRate(karakter legend[], int jumlahLegend) {
    for(int i =  0; i < jumlahLegend; i++){
        karakter key = legend[i];
        int j = i - 1;
        while(j >= 0 && legend[j].win_rate < key.win_rate) {
            swap(legend[j + 1], legend[j]);
            j--;
        }
        legend[j + 1] = key;
    }
    cout << "\nLegend berhasil diurutkan berdasarkan Win Rate tertinggi!\n";
    pause();
}
```

```
void searchid(karakter Legend[], int jumlahLegend){
    int cari_id;
    cout << "\nMasukkan ID legend yang ingin dicari: ";
    cin >> cari_id;

    int low = 0;
    int high = jumlahLegend - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (Legend[mid].id == cari_id) {
            cout << "\nLegend ditemukan:\n";
            cout << "ID: " << Legend[mid].id << "\n";
            cout << "Nama: " << Legend[mid].nama << "\n";
            cout << "Class: " << Legend[mid].classs << "\n";
            cout << "Best Weapon: " << Legend[mid].best_weapon << "\n";
            cout << "Pick Rate: " << Legend[mid].pick_rate << "%\n";
            cout << "Win Rate: " << Legend[mid].win_rate << "%\n";
            pause();
            return;
        } else if (Legend[mid].id < cari_id) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    cout << "\nID Legend tidak ditemukan.\n";
    pause();
}

void searchnama(karakter Legend[], int jumlahLegend){
    string cari_nama;
    cout << "\nMasukkan nama legend yang ingin dicari: ";
    cin >> cari_nama;

    for (int i = 0; i < jumlahLegend; i++) {
        if (Legend[i].nama == cari_nama) {
            cout << "\nLegend ditemukan:\n";
            cout << "ID: " << Legend[i].id << "\n";
            cout << "Nama: " << Legend[i].nama << "\n";
            cout << "Class: " << Legend[i].classs << "\n";
            cout << "Best Weapon: " << Legend[i].best_weapon << "\n";
            cout << "Pick Rate: " << Legend[i].pick_rate << "%\n";
            cout << "Win Rate: " << Legend[i].win_rate << "%\n";
            pause();
            return;
        }
    }
    cout << "\nLegend tidak ditemukan.\n";
    pause();
}
```

```cpp
do {
    menu_login();
    cin >> pilihan;

    switch (pilihan) {
        case 1: {
            indexUser = LoginUser(dataUser, &jumlahUser);
            if(indexUser != -1){
                cout << "\nLogin berhasil! Selamat datang, "
                    << dataUser[indexUser].usn << " (" << dataUser[indexUser].role << ")\n";
                    pause();

                    if (dataUser[indexUser].role == "admin") {
                        int menu_admin;
                        do{
                            menuAdmin();
                            cin >> menu_admin;
                            switch (menu_admin) {
                                case 1:
                                    TambahLegend(legend, &jumlahLegend);
                                break;

                                case 2:
                                    DaftarLegend(legend, jumlahLegend);
                                    break;

                                case 3:
                                    EditLegend(legend, &jumlahLegend);
                                    break;

                                case 4:
                                    HapusLegend(legend, &jumlahLegend);
                                    break;

                                case 5:
                                    sortClass(legend, jumlahLegend);
                                    break;

                                case 6:
                                    sortPickrate(legend, jumlahLegend);
                                    break;

                                case 7:
                                    sortWinRate(legend, jumlahLegend);
                                    break;

                                case 8:
                                    searchid(legend, jumlahLegend);
                                    break;

                                case 9:
                                    searchnama(legend, jumlahLegend);
                                    break;

                                case 0:
                                    logout();
                                    break;
                                default:
                                    cout << "Pilihan tidak valid. Silakan coba lagi.\n";
                                    pause();
                            }
                        } while (menu_admin != 0);
                    }
```

```
                        else if(dataUser[indexUser].role == "user") {
                            int menu_user;
                            do {
                                menuUser();
                                cin >> menu_user;
                                switch (menu_user) {
                                    case 1:
                                        DaftarLegend(legend, jumlahLegend);
                                        break;

                                    case 2:
                                        sortClass(legend, jumlahLegend);
                                        break;

                                    case 3:
                                        sortPickrate(legend, jumlahLegend);
                                        break;

                                    case 4:
                                        sortWinRate(legend, jumlahLegend);
                                        break;

                                    case 5:
                                        searchid(legend, jumlahLegend);
                                        break;

                                    case 6:
                                        searchnama(legend, jumlahLegend);
                                        break;

                                    case 7:
                                        BuatTier(tier_list, &jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 8:
                                        LihatTIer(tier_list, jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 9:
                                        EditTier(tier_list, &jumlahtier, legend, jumlahLegend);
                                        break;

                                    case 10:
                                        HapusTier(tier_list, &jumlahtier, legend, jumlahLegend);
                                        break;
                                    case 0:
                                        logout();
                                        break;

                                    default:
                                        cout << "Pilihan tidak valid. Silakan coba lagi\n";
                                        pause();
                                }
                            } while (menu_user != 0);
                        }
                        break;
                    }
                }

                if (indexUser == -1) {
                    percobaan--;
                    cout << "Login gagal! Sisa percobaan: " << percobaan << "\n";
                    pause();
                    if (percobaan == 0) {
                        cout << endl;
                        cout << "Anda telah kehabisan percobaan login. Program SELESAI.\n";
                        cout << endl;
                        exit(0);
                    }
                }
                break;
            case 2: {
                RegisterUser(dataUser, &jumlahUser);
                break;
            }
            case 0:
                cout << endl;
                cout << "Terima kasih telah menggunakan program ini. Sampai jumpa!\n";
                cout << endl;
                break;
            default:
                cout << "Pilihan tidak valid. Silakan coba lagi.\n";
                pause();
        }
    } while (pilihan != 0);

    return 0;
}
```

```
void TambahLegend(karakter legend[], int *jumlahLegend) {
    cout << "\nTambah Legend\n";
    cout << "Nama: ";
    cin >> legend[*jumlahLegend].nama;
    cout << "Class: ";
    cin >> legend[*jumlahLegend].classs;
    cout << "Best Weapon: ";
    cin >> legend[*jumlahLegend].best_weapon;
    while(true){
    try{
        cout<<"Pick Rate: ";
        cin>>legend[*jumlahLegend].pick_rate;

        if(cin.fail()) throw invalid_argument("\nHarus angka!\n");
        if(legend[*jumlahLegend].pick_rate <0 || legend[*jumlahLegend].pick_rate >100)
            throw invalid_argument("\nharus dibawah 100!\n");

        break;
    }
    catch(const invalid_argument& e){
        cout<<e.what()<<endl;
        cin.clear();
        cin.ignore(1000,'\n');
    }
}
    while(true){
    try{
        cout<<"Win Rate: ";
        cin>>legend[*jumlahLegend].win_rate;

        if(cin.fail()) throw invalid_argument("\nHarus angka!\n");
        if(legend[*jumlahLegend].win_rate <0 || legend[*jumlahLegend].win_rate >100)
            throw invalid_argument("\nharus dibawah 100!\n");

        break;
    }
    catch(const invalid_argument& e){
        cout<<e.what()<<endl;
        cin.clear();
        cin.ignore(1000,'\n');
    }
}
    (*jumlahLegend)++;
    cout << "Legend berhasil ditambahkan!\n";
    pause();
}
```

```
void Editlegend(karakter legend[], int &jumlahlegend) {
    cout << "\nEdit legend\n";
    if (=jumlahlegend == 0) {
        cout << "Belum ada legend yang dapat diedit.\n";
        pause();
    } else {
        cout << "\n Data Legend \n";
        cout << "|" << setw(10) << left << "Nama" << "|" << setw(10) << left << "Class" << "|" << setw(10) << left << "Best Weapon" << "|" << setw(10) << left << " Pick Rate" << "|" << setw(10) << left << " Win Rate |\n";
        cout << "==========================================================\n";
    for (int i = 0; i < =jumlahlegend; i++) {
        cout << "| " << setw(10) << left << legend[i].nama << " | " << setw(10) << left << legend[i].classs << " | " << setw(10) << left << legend[i].best_weapon << " | " << setw(8) << left << legend[i].pick_rate << setw(5)<< "%" << " | " << setw(8) << left << legend[i].win_rate << "% |"  << "\n";
    }
    cout << "==========================================================\n";
        int index;
        cout << "\nPilih nomer legend yang ingin diedit: ";
        cin >> index;
        if (index < 1 || index > =jumlahlegend) {
            cout << "Pilihan tidak valid.\n";
        } else {
            index --;
            cout << "\nEdit legend: " << legend[index].nama << "\n";
            cout << "\nNama: ";
            cin >> legend[index].nama;
            cout << "Class: ";
            cin >> legend[index].classs;
            cout << "Best Weapon: ";
            cin >> legend[index].best_weapon;
            while(true){
            try{
                cout<<"Pick Rate: ";
                cin>>legend[index].pick_rate;

                if(cin.fail()) throw invalid_argument("\nHarus angka!\n");
                if(legend[index].pick_rate <0 || legend[index].pick_rate >100)
                    throw invalid_argument("\nharus dibawah 100!!\n");
            break;
            }
            catch(exrat invalid_argument& e){
            cout<<e.what()<<endl;
            cin.clear();
            cin.ignore(1000,'\n');
    }
}

            while(true){
            try{
                cout<<"Win Rate: ";
                cin>>legend[index].win_rate;

                if(cin.fail()) throw invalid_argument("\nHarus angka!\n");
                if(legend[index].win_rate <0 || legend[index].win_rate >100)
                    throw invalid_argument("\nharus dibawah 100!!\n");

            break;
    }

            catch(exrat invalid_argument& e){
                cout<<e.what()<<endl;
                cin.clear();
                cin.ignore(1000,'\n');
    }
}

            cout << "\nLegend berhasil diedit!\n";
            pause();
        }
    }
}
```

4. Hasil Output

```
===== MENU LOGIN =====
1. Login
2. Register
3. Keluar
Pilih menu:
```

```
Pilih menu: 2
Masukkan Username: Cordova
Masukkan Password: 123
Masukkan role (admin/user): user
Selamat Registrasi berhasil! Silahkan Login
```

```
Pilih menu: 1

Masukkan Username: admin
Masukkan Password: admin123

Login berhasil! Selamat datang, admin (admin)
```

Gambar 4.1 Tampilan Login

```
===== MENU ADMIN =====
1. Tambah Legend
2. Lihat Daftar legend
3. Edit Legend
4. Hapus Legend
5. Logout
Pilih menu:
```

```
Pilih menu: 1

Tambah Legend
Nama: Octane
Class: Skimisher
Best Weapon: SMG
Pick Rate: 20.4
Win Rate: 3.7
Legend berhasil ditambahkan!
```

```
 Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
=================================================================
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
| Octane     | Skimisher  | SMG        | 20.4%     | 3.7% |
=================================================================
```

```
 Data Legend
|Nama         |Class        |Best Weapon | Pick Rate| Win Rate |
=================================================================
| Alter       | Skirmisher | SMG         | 14.5%      | 52% |
| Ash         | Skirmisher | SMG         | 0.4%      | 49.4% |
| Bangalore   | Assault    | SMG         | 7.4%      | 49.1% |
| Fuse        | Assault    | SMG         | 1.1%      | 48.8% |
| Horizon     | Skirmisher | AR          | 3.6%      | 47.5% |
| Loba        | Support    | AR          | 2.8%      | 51.7% |
| Newcastle   | Support    | SG          | 0.6%      | 48.1% |
| Seer        | Recon      | AR          | 1.3%      | 48.7% |
| valkyrie    | Recon      | AR          | 9.2%      | 54.4% |
| Wattson     | Controller | SMG         | 0.2%      | 51.7% |
| Wraith      | Skirmisher | SMG         | 0.6%      | 47.5% |
| Octane      | Skimisher  | SMG         | 20.4%      | 3.7% |
=================================================================

Pilih nomor legend yang ingin diedit: 12

Edit Legend: Octane

Nama: Octane
Class: Skimisher
Best Weapon: SG
Pick Rate: 30.2
Win Rate: 51.4

Legend berhasil diedit!
```

```
 Data Legend
|Nama         |Class        |Best Weapon | Pick Rate| Win Rate |
=================================================================
| Alter       | Skirmisher | SMG         | 14.5%      | 52% |
| Ash         | Skirmisher | SMG         | 0.4%      | 49.4% |
| Bangalore   | Assault    | SMG         | 7.4%      | 49.1% |
| Fuse        | Assault    | SMG         | 1.1%      | 48.8% |
| Horizon     | Skirmisher | AR          | 3.6%      | 47.5% |
| Loba        | Support    | AR          | 2.8%      | 51.7% |
| Newcastle   | Support    | SG          | 0.6%      | 48.1% |
| Seer        | Recon      | AR          | 1.3%      | 48.7% |
| valkyrie    | Recon      | AR          | 9.2%      | 54.4% |
| Wattson     | Controller | SMG         | 0.2%      | 51.7% |
| Wraith      | Skirmisher | SMG         | 0.6%      | 47.5% |
| Octane      | Skimisher  | SG          | 30.2%      | 51.4% |
=================================================================

Pilih nomor legend yang ingin dihapus: 1

Legend berhasil dihapus!
```

```
===== MENU ADMIN =====
1. Tambah Legend
2. Lihat Daftar legend
3. Edit Legend
4. Hapus Legend
5. Logout
Pilih menu: 5
Logout berhasil. Kembali ke menu utama.
```

Gambar 4.2 Tampilan Menu dan Fitur Admin

```
Masukkan Username: Cordova
Masukkan Password: 123

Login berhasil! Selamat datang, Cordova (user)

===== MENU USER =====
1. Lihat Daftar Legend
2. Buat Tier List
3. Lihat Tier List
4. Edit Tier List
5. Hapus Tier List
6. Logout
Pilih menu:
```

```
Buat Tier List
Masukkan judul tier list: Tier list Ultimate Legend

Masukkan tier setiap legend (S/A/B/C/D):
Tier legend Ash  : S
Tier legend Bangalore: S
Tier legend Fuse : A
Tier legend Horizon: B
Tier legend Loba : B
Tier legend Newcastle: A
Tier legend Seer : C
Tier legend valkyrie: S
Tier legend Wattson: D
Tier legend Wraith: B
Tier legend Octane: A
Tier list berhasil dibuat!
```

```
Pilih menu: 3

Lihat Tier List
1. Tier list Ultimate Legend

Pilih tier list yang ingin dilihat: 1

 Tier List: Tier list Ultimate Legend
Ash         = S
Bangalore   = S
Fuse        = A
Horizon     = B
Loba        = B
Newcastle   = A
Seer        = C
valkyrie    = S
Wattson     = D
Wraith      = B
Octane      = A
```

```
Pilih menu: 4

Daftar Tier List.
1. Tier list Ultimate Legend

Pilih tier list yang ingin diedit: 1

Masukkan tier baru setiap legend (S/A/B/C/D):
Masukkan tier Baru Ash:    C
Masukkan tier Baru Bangalore:    A
Masukkan tier Baru Fuse:    S
Masukkan tier Baru Horizon:    A
Masukkan tier Baru Loba:    A
Masukkan tier Baru Newcastle:    B
Masukkan tier Baru Seer:    C
Masukkan tier Baru valkyrie:    S
Masukkan tier Baru Wattson:    B
Masukkan tier Baru Wraith:    A
Masukkan tier Baru Octane:    B

Tier list berhasil diedit
```

```
Pilih menu: 5

Daftar Tier List.
1. Tier list Ultimate Legend

Pilih tier list yang ingin dihapus: 1

Tier list berhasil dihapus
```

Gambar 4.3 Tampilan Menu dan Fitur User

```
===== MENU LOGIN =====
1. Login
2. Register
3. Keluar
Pilih menu: 3

Terima kasih telah menggunakan program ini. Sampai jumpa!
```

Gambar 4.4 Tampilan Keluar Program

```
Pilih menu: 5

Legend berhasil diurutkan berdasarkan Class!

Tekan Enter untuk melanjutkan...
```

```
 Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
==============================================================
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
==============================================================
```

Gambar 4.5.1 Tampilan Sorting berdasarkan Class (A-Z)

```
Pilih menu: 6

Legend berhasil diurutkan berdasarkan Pick Rate tertinggi!

Tekan Enter untuk melanjutkan...
```

```
Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
==============================================================
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
==============================================================
```

Gambar 4.5.2 Tampilan Sorting berdasarkan Pic Rate Tertinggi

```
Pilih menu: 7

Legend berhasil diurutkan berdasarkan Win Rate tertinggi!

Tekan Enter untuk melanjutkan...
```

```
 Data Legend
|Nama        |Class       |Best Weapon | Pick Rate| Win Rate |
==============================================================
| valkyrie   | Recon      | AR         | 9.2%     | 54.4% |
| Alter      | Skirmisher | SMG        | 14.5%     | 52% |
| Loba       | Support    | AR         | 2.8%     | 51.7% |
| Wattson    | Controller | SMG        | 0.2%     | 51.7% |
| Ash        | Skirmisher | SMG        | 0.4%     | 49.4% |
| Bangalore  | Assault    | SMG        | 7.4%     | 49.1% |
| Fuse       | Assault    | SMG        | 1.1%     | 48.8% |
| Seer       | Recon      | AR         | 1.3%     | 48.7% |
| Newcastle  | Support    | SG         | 0.6%     | 48.1% |
| Horizon    | Skirmisher | AR         | 3.6%     | 47.5% |
| Wraith     | Skirmisher | SMG        | 0.6%     | 47.5% |
==============================================================
```

Gambar 4.5.3 Tampilan Sorting berdasarkan Win Rate Tertinggi

```
Pilih menu: 8

Masukkan ID legend yang ingin dicari: 105

Legend ditemukan:
ID: 105
Nama: Horizon
Class: Skirmisher
Best Weapon: AR
Pick Rate: 3.6%
Win Rate: 47.5%
```

```
Masukkan ID legend yang ingin dicari: 1019

ID Legend tidak ditemukan.
```

Gambar 4.6.1 Tampilan Searching ID

```
Masukkan nama legend yang ingin dicari: Wraith

Legend ditemukan:
ID: 1011
Nama: Wraith
Class: Skirmisher
Best Weapon: SMG
Pick Rate: 0.6%
Win Rate: 47.5%
```

```
Masukkan nama legend yang ingin dicari: warit

Legend tidak ditemukan.
```

Gambar 4.6.2 Tampilan Searching Nama Legend

5. Langkah-langkah GIT

5.1 GIT Add

berfungsi untuk menambahkan file atau perubahan ke staging area, sebelum disimpan permanen

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git add post-test
```

5.2 GIT Commit

befungsi untuk menyimpan perubahan yang sudah ada di staging area ke dalam repository Git beserta pesan (message) sebagai catatan perubahan

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git commit -m "up pt 7 apl"
[main 4ecae16] up pt 7 apl
 2 files changed, 66 insertions(+), 8 deletions(-)
```

5.3 GIT Push

berfungsi untuk mengirim atau (upload) commit dari repository local ke repository Github

```
PS D:\Tugas Kuliah\Tugas Praktikum\praktikum-apl> git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 221.19 KiB | 2.28 MiB/s, done.
Total 6 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
To https://github.com/Cordova88/praktikum-apl.git
   d922a88..4ecae16  main -> main
```